



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2

2025/2026

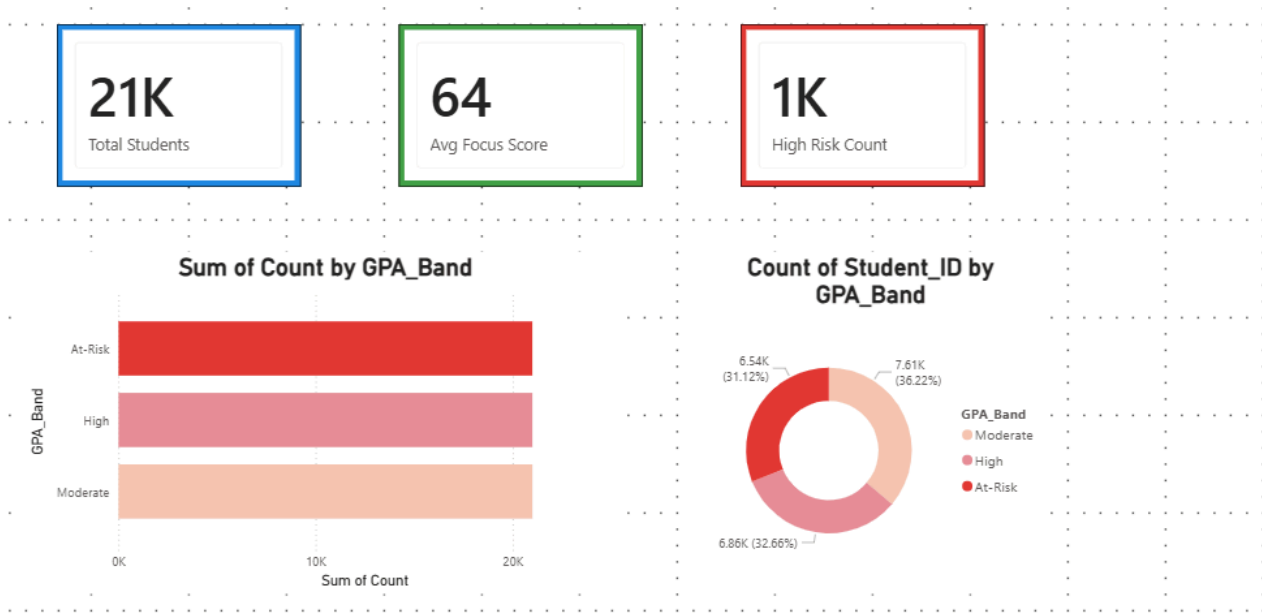
SECP3213-02 BUSINESS INTELLIGENCE

TOPIC: VISUALIZATION & DASHBOARD

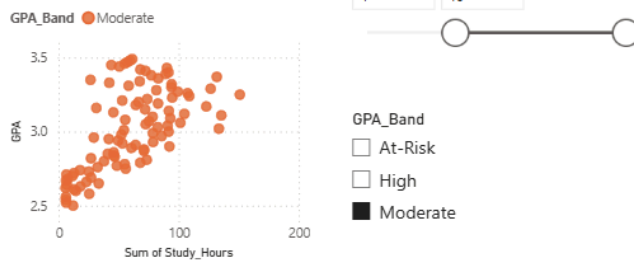
NAME	MATRIC NUMBER
DAYANG FARAH FARZANA BINTI ABANG IDHAM	A23CS0071
FARRA NURZAHIN BINTI ZAHARIL ANUAR	A23CS0079
SAFIYA NURSYAHADAH BINTI MASNOOR	A23CS0176

LECTURER: DR. NOORFA HASZLINNA BINTI MUSTAFFA

1.0 Screenshots of Dashboard



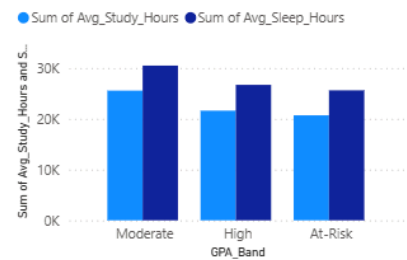
How Study Hours Directly Drive GPA Tiers



Explanation: Maps student study hours against raw GPA scores to track the exact academic return on study time.

Key Insight: More study hours lead directly to a higher GPA, but high stress scores break this pattern and drag grades down.

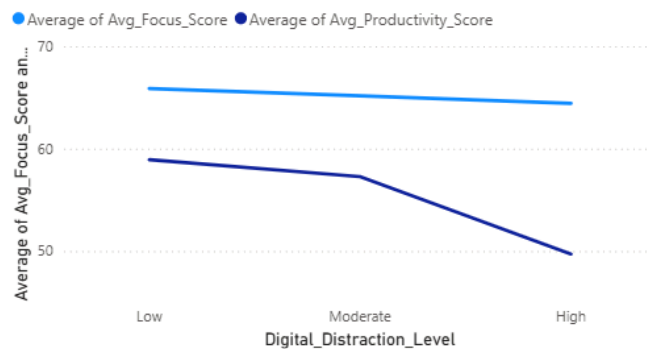
Comparing Study vs. Sleep Dedication Across GPA Bands



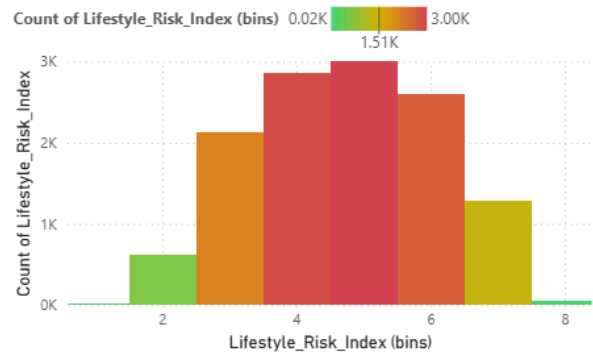
Explanation: Compares the average weekly study hours and average sleep hours side-by-side across the three GPA bands.

Key Insight: High-performing students maintain a balanced 1:1 ratio of sleep and study, whereas at-risk students show a massive drop in study hours but do not get extra sleep.

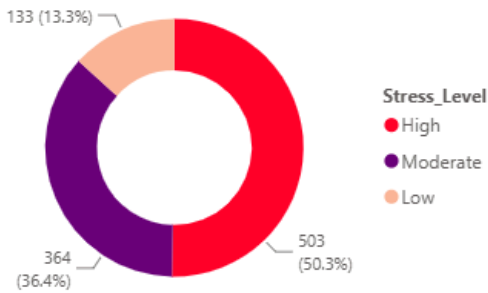
Performance Trends by Distraction Level



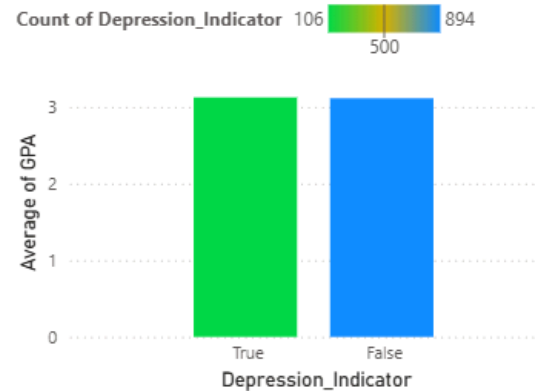
Student Count by Risk Category



Count of Student_ID by Stress_Level



Average Academic GPA by Depression Indicator



2.0 Key Insights

- (a) **Direct Return** : There is a clear, statistically significant correlation where increasing weekly study hours leads to higher GPA tiers. This establishes that dedicated study time is a primary, predictable driver of academic achievement. Students who invest more time consistently see higher performance outcomes.
- (b) **Stress Overlap** : High psychological stress scores disrupt the positive correlation between study time and GPA. Even with high study hours, excessive stress acts as a destructive variable that causes grades to fluctuate or fall. This reveals that academic engagement is often ineffective without concurrent stress management.
- (c) **The Attention Tax**: There is a clear, downward trend showing that as digital distraction levels escalate from Low to High, both cognitive focus and operational productivity scores experience a significant, parallel drop. This establishes that unmanaged digital habits impose a direct "attention tax" on students, systematically eroding the mental clarity and efficiency required to handle academic workloads.
- (d) **Mainstream Lifestyle Risk**: The distribution density reveals that the vast majority of the student population is heavily concentrated within the moderate-to-high risk categories (bins 4 through 6). This proves that poor lifestyle routines and high-risk digital behaviors are not isolated anomalies confined to a small group of failing students but they represent a widespread, mainstream campus culture that requires broad institutional intervention.

- (e) **The High-Functioning Mask:** The data reveals an identical average GPA of **3.11 to 3.12** across both depressed and non-depressed student cohorts. This indicates a profound "high-functioning" academic survival trend where students facing significant mental health struggles continue to preserve their grades at the expense of their internal well-being. Academic performance alone can no longer be used by advisors as a proxy or warning sign for mental health issues.
- (f) **Pervasive Academic Pressure:** Over **86%** of the student body falls into either the High or Moderate stress categories. This overwhelming concentration proves that psychological strain is a systemic institutional crisis rather than an isolated individual issue, directly tying universal academic pressure to the broader campus climate.
- (g) **Quantifiable Institutional Risk:** The executive KPI metrics reveal that approximately 1,000 students out of a total population of 21,000 have breached the critical high lifestyle risk threshold, representing a statistically significant 4.76% of the entire student body. This concentration of at-risk individuals within a single institution confirms that deteriorating lifestyle habits constitute a measurable, systemic liability rather than an isolated behavioural anomaly, warranting immediate structured intervention at the university governance level.
- (h) **Chronic Focus Deficiency:** The global average focus score of 64 recorded across the entire student population falls substantially below optimal cognitive performance benchmarks, signalling an institution-wide attention deficit that transcends individual academic disciplines or year groups. This pervasive decline in student concentration capacity is not an independent variable but rather a downstream consequence of compounding lifestyle risk factors including excessive digital engagement, insufficient sleep, and elevated stress levels as evidenced by the broader dashboard trends identified across all four analytical dimensions.

3.0 Business Recommendations

(a) **Implement a "Predictive Academic Alert System" and Target 40-Hour Peer Workshops:**

Institutional advisors should deploy an automated tracking system to monitor weekly study engagement and flag students dropping below the critical 40-hour threshold before exams occur. To support these flagged individuals, the university should establish peer-led time management workshops that teach at-risk students how to structure their schedules to achieve a balanced 1:1 study-to-sleep ratio. Combining structural tracking with stress-reduction counseling will effectively patch lifestyle leaks and safeguard student retention rates.

(b) **Create Phone-Free Study Areas and Launch Digital Habit Campaigns:** To directly fix high digital distraction scores, university management should convert specific sections of the campus library into dedicated phone-free zones. In these spaces, students will be required to store their smartphones in secure lockboxes while they study to prevent screen multitasking. Alongside these physical spaces, the student welfare team should launch simple digital wellness campaigns that share practical tips on avoiding social media habits. Providing these quiet, interruption-free environments will help students break the cycle of screen fragmentation, directly recovering their lost focus and boosting daily study productivity.

(c) **Start Universal Mental Health Screenings and Support High-Achieving Students:** Because students facing depression still manage to keep their grades steady, academic advisors cannot simply look at dropping marks to see who needs help. Instead, the university should introduce quick, private health surveys that every student completes during routine class registration each semester. Checking in on everyone early will allow counselors to find and assist at-risk individuals before a serious mental health crisis happens, rather than waiting for their academic performance to drop.

(d) **Institutionalise a Real-Time Student Wellness KPI Governance Framework for University Management:** Senior university leadership should mandate the deployment of a live institutional dashboard continuously monitoring population-level wellness KPIs, including total high-risk student volume and global average focus score, across each semester cycle. Embedding these metrics into formal governance reporting will shift institutional decision-making from reactive crisis management toward proactive, data-driven student welfare stewardship, with a target of

reducing the high-risk population below 3% within two consecutive academic years as a measurable accountability benchmark.

- (e) **Integrate Mandatory Lifestyle Risk Screening into Semester Registration Protocols for Student Welfare Units:** Student Welfare Units should embed a standardised lifestyle risk assessment directly into the university's semester re-registration process to ensure universal screening coverage across the entire student population. Given that empirical evidence identifies approximately 1 in 21 students already operating above the critical risk threshold, early systematic identification will enable welfare counsellors to intervene before lifestyle deterioration escalates into academic failure or mental health crisis, repositioning the welfare unit's role from reactive resolution to preventative student care.